LEARNER GUIDE

Meet the parabola

Quadratics. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You have met a quadratic as a growing sequence, as a rule, and as an expression. Today you meet its third face: a shape. The rule is the simplest quadratic of all, $y = x^2$ — "the height is the input, squared". To draw it you do exactly what you already know: substitute an x, work out y, and plot the point (x, y). Here is the surprise that makes the shape. Squaring a negative number gives a positive answer: $(-3)^2 = 9$, the same as $3^2 = 9$. So the negative side of the table climbs back up to match the positive side, and the points fall into a symmetric U. The line down the middle, where the curve turns, is its line of symmetry, and the lowest point, (0, 0), is its turning point. That U-shaped curve has a name: a parabola. You will build it from a table, predict heights and watch them land on the curve, find mirror twins across the line of symmetry, and — as a stretch — plot half-way points to see that the curve is smooth everywhere. There is no calculator today. The question you are exploring is: why does $y = x^2$ bend into a U?

What you are learning to notice

- That $y = x^2$ is a rule: substitute an x, square it, and you have the height y.
- That squaring a negative gives a positive, so the two arms of the curve rise the same way.
- That the points make a symmetric U — a parabola — with a line of symmetry down $x = 0$ and a turning point at (0, 0).
- That mirror pairs (such as –3 and 3) share the same height, so knowing one arm tells you the other.

Moving through it, one step at a time

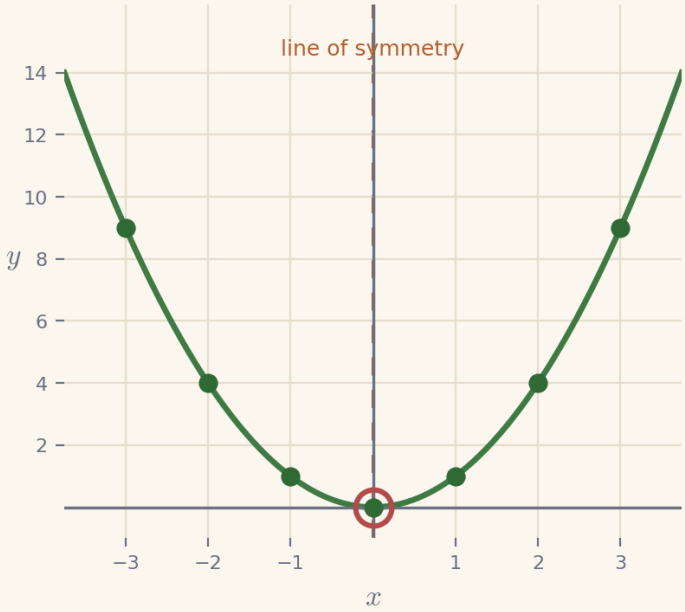
In Build the curve from a table, fill in $y = x^2$ for each x from –3 to 3, plot the points, then join them into a smooth curve. In Predict the height, choose an x, work out $y = x^2$ and watch the point land on the curve. In The mirror, find the twin of a given point across the line of symmetry. In Between the whole numbers (a stretch), plot a half-way x such as 1.5 and see it land neatly on the curve between the dots.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — build the parabola from a table

Make a table for $y = x^2$, plot the points, and join them. Substitute each x and square it.

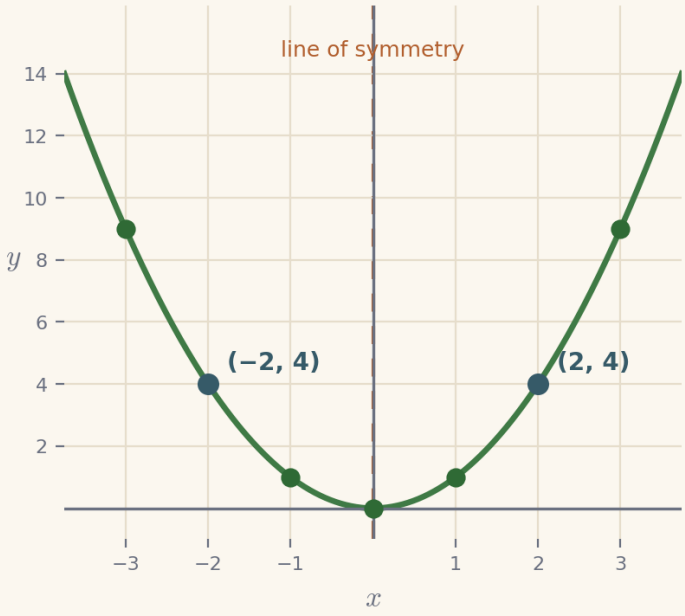


1. $x = -3$ $(-3)^2 = 9$
2. $x = -1$ $(-1)^2 = 1$
3. $x = 0$ $0^2 = 0$
4. $x = 2$ $2^2 = 4$

the seven points make a symmetric U — a parabola — with line of symmetry $x = 0$ and turning point (0, 0)

Example 2 — find the mirror twin

The point (2, 4) is on $y = x^2$. Find its mirror twin — the other point at the same height.



1. the point given	(2, 4)
2. its height is	$2^2 = 4$
3. a twin at height 4 needs	$x^2 = 4$
4. the other value is	$x = -2$, since $(-2)^2 = 4$

the mirror twin is (–2, 4); the two mirror across the line of symmetry $x = 0$

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substituting into a rule, squaring a number including squaring a negative, and plotting coordinates), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.